Table 7.1 The expected long-time computed average using the given second-order Langevin dynamics discretization scheme

Scheme	$\langle q^2\rangle_h/\langle q^2\rangle$	$\langle p^2\rangle_h/\langle p^2\rangle$	$\langle qp\rangle$
⟦ABOBA⟧	1	$[1-h^2\Omega^2/(4m)]^{-1}$	0
⟦OBABO⟧	$[1-h^2\Omega^2/(4m)]^{-1}$	1	0
⟦BAOAB⟧	1	$1-h^2\Omega^2/(4m)$	0
⟦OABAO⟧	$1-h^2\Omega^2/(4m)$	1	0

schemes, we are able to compute the exact expected average by solving the linear equations. The results are summarized in Table 7.1, where we have not considered the ⟦AOBOA⟧ and ⟦BOAOB⟧ schemes due to their inconsistency at large friction (this does not preclude the analysis, but limits each those method's usefulness in simulation). Similar analysis can be conducted for more complicated composition schemes involving multiple force evaluations.

The mentioned splitting schemes give a second-order error in the computed averages, but surprisingly each gives an exact result for computed averages of either position or momentum. The exactness property is not born out of the symmetry of the method, but rather of the specific order of terms and choice of splitting "building blocks". Even non-symmetric first-order schemes can exhibit similar qualities. For example, analysis shows that the schemes denoted ⟦AOAB⟧ or ⟦BAOA⟧ both exhibit a first order error in the covariance $\langle qp\rangle_h = \mathcal{O}(h)$, however they give exact results for both $\langle q^2\rangle_h$ and $\langle p^2\rangle_h$.

7.4.2 More General Schemes

We can conduct a similar analysis for any integrator that we can express in the form (7.7), simply by plugging in the relevant constants and solving a linear equation. The difficulty lies when random numbers are reused, leading to correlation between steps. This requires careful treatment to compute the expectation.

Example 7.2 Consider the variant of the BBK integrator that recycles the random number between steps:

Brünger-Brooks-Karplus with post-step re-use of random number (BBK*)

$$\boldsymbol{p}_{n+1/2} = (1-h\gamma/2)\boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_n$$
$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}$$
$$\boldsymbol{p}_{n+1} = [\boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_{n+1}]/(1+h\gamma/2),$$

where $\mathbf{R}_n$ is a vector of normal random numbers with zero mean and unit variance. In some systems, it can be computationally expensive to generate normal random numbers for each degree of freedom, hence algorithms may re-use the random